

if available, flip cells or select suitable alternatives, if available in the library. IC Compiler will use colour information available in the library to ensure compliance. IC Validator in Design, which has a foundry-certified colouring engine, is available within the IC Compiler environment to perform a fast check, if needed, during the implementation process.

The router needs to provide a two-colourable solution at the end of design while minimising the impact on area and runtime. The key to this is not just applying the DPT spacing rules blindly, which would ensure DPT correctness but also lead to area explosion. Zroute, the router within IC Compiler, utilises multiple techniques to ensure DPT readiness and applies them intelligently. One of the techniques utilised by the router is selective use of spacing rules and knowing when to apply which rules and when to apply other techniques. The second technique looks at routed patterns and ensures that there are no odd cycles or DPT violations created so that every route can go on an alternate mask.

The Synopsys Custom Designer tool for mixed-signal design provides a design-rule-driven layout capability that supports the new 20nm rules including spacing rules and new complex via requirements. This capability will significantly help designers of custom circuits comply.

2 In-design physical verification

20 nm involves in-design physical verification throughout the flow to reduce time-consuming, uncertain iterations. It is interesting to see what is being done in this area.

To provide designers with a complete end-to-end design and sign-off solution, Cadence integrates its physical verification system with industry-standard Cadence Virtuoso custom/mixed-signal and Encounter digital design flows, informs Ahuja.

The physical verification system is a trusted solution to enable users to achieve advanced node design sign-off.

It is efficient in reducing debug time, thus increasing productivity. The solution supports double patterning, advanced device extraction and 3D IC. It extends physical verification technology into design reliability checking and constraint validation. It also offers a distributed multi-threading processing capability and greatly accelerates throughput without requiring specialised hardware.

Biddle says, "Our experience with 20nm has shown this to be a real concern. Finding DPT violations late in the flow can have a significant effect on the design closure time. In addition, resolving DPT violation manually will be very time-consuming. A solution may not be obvious on looking at the violation."

The Synopsys IC Validator incorporates a comprehensive range of techniques using a combination of property and equation based techniques to check conformity to foundry rules. By incorporating IC validation in-design, it provides a unified environment where the designer can avoid the potentially huge time lost in addressing issues like DPT rule violations detected late in the design process.

While many of the double-patterning rules could be built into place-and-route tools, implementation tools may not be able to see conflicts or rule violations at chip level or across hierarchical boundaries. Synopsys recommends fixing of physical violations at each step in the flow. IC Validator In-Design works natively with IC Compiler to ensure that these checks with many complex rules can be made very quickly to converge on sign-off closure efficiently.

A new, native hierarchical DPT colouring engine in IC Validator enables fast DPT decomposition analysis. The colouring engine makes it possible to accurately detect and quickly resolve DPT colour conflicts. IC Validator provides a mechanism for automatic detection and repair of DPT violation. In addition, the resultant GDSII for the chip will not be coloured, enabling the foundries to make optimal final decomposition choices. IC Validator's in-design technology provides colouring

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